Department of Electrical Engineering

Indian Institute of Technology Madras

EE 2005: Electrical Machines

Tutorial Sheet on 3 Phase transformers and p.u

1. A 3-phase transformer is used to step down the voltage from 3300 V supply mains. The turns ratio per phase of the transformer is 10. If the load is adjusted such that it draws 10 A of current from the supply mains in either of the two cases below, determine the secondary line voltage, line current and output kVA when the transformer windings are connected in (a) star-delta (b) delta-star.

   Y/Δ: 190.526V, 173.21A, 57.15 kVA, Δ/Y : 571.576 V, 57.73A, 57.15 kVA

2. A 100 MVA, 230/115 kV delta – delta connected 3 phase power transformer has a resistance of 0.02 p.u and a reactance of 0.055 pu. Find the voltage regulation of this transformer when it is supplying an inductive load of 80 MVA at 115 kV and 0.85 power factor.

   input voltage is 3.718% higher

3. A 150 kVA 3 phase transformer is rated 2400/460 V and has a series impedance of 0.038 + j 0.135 p.u on its own base. It supplies power to a 3-phase balanced load drawing 95 kVA at unity power factor and at a terminal line voltage of 438 V. Calculate the line voltage at the HV side of the transformer. Do the calculations on a 460 V, 100 kVA base. 2355.74 V

4. A delta-star connected bank of three identical 100 kVA 2400/120 V, 50 Hz transformers is supplied with power through a feeder whose impedance is 0.3+j0.8 Ω per phase. The voltage at the sending end of the feeder is held constant at 2400 V line to line. The results of a single-phase short circuit test on one of the transformers with its low-voltage terminals short-circuited are V_H=57.5 V, f=50 Hz, I_H=41.6 A, P=875 W.
   a. Determine the secondary line to line voltage when the bank delivers rated current to a balanced three phase unity pf load.
   b. Compute the currents in the transformer primary and secondary windings and in the feeder wires if a solid three phase short circuit occurs at the secondary line terminals.

5. Three Single phase transformers each rated 25 kVA, 3300/300 V are delta – delta connected to form a 3 phase bank. It is used to supply a load of 50 kVA at the secondary side. (a) Calculate the current in the primary winding of each transformer. During operation, one of the transformers develops a fault and is removed from service, with the remaining transformers used to supply the load in an open delta configuration. (a) Calculate the kVA supplied by each transformer and the total kVA rating of the set. (b) If the load has a power factor of 0.95 lagging, calculate the power factor of operation of each transformer. What would be the power factor when the load is purely resistive?

****